Exercise 3: reproducible gauge-transformation experiments

Abstract

The question is whether two visibly different vector potentials represent the same uniform magnetic field. The supplied exercise gives A=(0,bx,0) and A′=(-by,0,0). Their difference is the gradient of s=-bxy+C; both curls equal (0,0,b). An interpolating family Aλ=A+λ∇s preserves this curl and all closed-loop circulation. Exact differentiation establishes the result; numerical experiments and browser checks test whether the implementation represents it faithfully.

Methodology and evidence

We transcribed the screenshot, derived the connecting scalar, tested the mathematics before implementation, and reused one sampled grid in both renderers. The experiments below document the hypothesis, evidence, problems, and corrections without treating finite numerical samples as a proof.

- Experiment 1: exploratory central-difference check.
- Experiment 2: test-driven physics and scientific exports.
- Experiment 3: native slider interaction and responsive rendering.
- Acceptance and review: completed checks, corrections, and explicit limits.

The PDF companion assembles this summary, the individual experiment records, and the generated four-panel figure from their existing sources. It is a local reproducibility report, not a claim of new experimental physics.

Conclusion and limitations

Gauge freedom changes the vector potential while preserving the magnetic field. Here b is constant and the gauge function is time-independent and smooth over all space. Coordinates and magnitudes use illustrative units. The plotted slice contains all spatial dependence because the fields have no z dependence. The browser checks cover Chromium; the interactive renderer needs access to its pinned D3 CDN. The scientific PNG/PDF and the algebra can be read offline.

The primary source is the supplied screenshot. Independent corroboration is MIT's electromagnetism gauge and circulation derivation, page 6.

Experiment 1 — independently check the proposed gauge family

Date: 2026-10-10 (Europe/Berlin).

Question and hypothesis

Does $A_\lambda=(-\lambda by,(1-\lambda)bx,0)$ have curl $(0,0,b)$ for every selected gauge? Differentiation predicts $(1-\lambda)b+\lambda b=b$. The corresponding gauge function is $s=-bxy$, and its constant offset is irrelevant.

Method

During planning, an in-memory Python calculation evaluated central differences with step 0.0001 at (-2,1), (0,0), and (1,-2). It combined $b=-1,0,1$ with $\lambda=0,0.25,0.5,0.75,1$, giving 45 point/gauge/strength cases. The test compared the measured z curl with b using absolute tolerance 1e-9. The implementation session's tool output is the evidence for this exploratory run; it did not create a raw log file or alter the repository.

Result and interpretation

All 45 cases passed. The maximum absolute error was approximately 1.56e-12. This small residual is floating-point subtraction error in the finite difference, not a physical change in the magnetic field. The analytic result remains exact. Subsequent grid tests use exactly representable sample spacing and are documented separately rather than retroactively replacing this result.

Learning and choice

The continuous family gives the slider a direct mathematical interpretation. Its midpoint is the symmetric gauge, making the screenshot's discussion of rotational symmetry visible. We chose this family because it connects the actual exercise endpoints without adding another physical model.

The mathematical convention and invariant closed-loop integral were checked against MIT's solution, page 6.

Experiment 2 — test the field model and inspect scientific figures

Date: 2026-10-10 (Europe/Berlin).

Question and hypothesis

Can one small implementation produce comparable plots without losing the exercise's field signs, gradient relation, or circulation? Shared coordinates, equal aspect, and a fixed arrow scale should preserve meaningful comparisons.

Method and evidence

We wrote endpoint, midpoint, scalar-gradient, curl, and circulation tests before implementing the model. The first test run failed during collection because the field module did not yet exist. After implementation, 22 tests passed. The production validation evaluates 15 gauge/strength combinations on a 17-by-17 grid and checks a counterclockwise unit square independently.

The first complete command finished at 11:55:41 UTC. Its timestamped generation log recorded zero maximum residuals for the grid curl, gradient identity, and unit-square circulation, with a tolerance of 1e-10. These affine/bilinear fields and dyadic sample spacing yield exact results in those particular floating-point calculations; this does not imply arbitrary numerical derivatives are exact. Each run saves current residuals in outputs/validation.json.

Issue, correction, and result

Visual inspection found small diagonal artifacts in the zero scalar contour near the saddle at the origin. The exact zero contour is the pair of coordinate axes. We therefore retained the exact axes and plotted only nonzero contour levels. This removes an ambiguous contour interpolation at a degenerate crossing without changing field data. The corrected pipeline completed at 11:56:14 UTC.

The PNG was inspected for arrows, sign changes, labels, contour geometry, and uniform perpendicular B symbols. The PDF is checked for one page and no raster images. The working mathematical core and its outputs were pushed in commit 84f11f8 before the interactive feature and final review.

Learning and choice

Matplotlib was sufficient; a custom scientific renderer would add unnecessary work. Its quiver documentation guided Cartesian arrow directions, explicit scale, and common pivot. Zero vectors are omitted. The separate B panel avoids projecting the entire magnetic field to zero in the xy plane.

Experiment 3 — verify interaction and responsive presentation

Date: 2026-10-10 (Europe/Berlin).

Question and hypothesis

Does moving the gauge slider visibly change A while leaving B unchanged, and does the presentation remain readable on narrow screens? A native range input and fixed sampled endpoints should support this without a backend.

Method

The generator embeds its existing endpoint samples in the browser view. Chromium opens the generated local file, uses End and Home keys to reach the exercise endpoints, and applies 50 ArrowRight presses to reach λ=0.50. The check compares all potential arrow paths and the complete B-symbol group. It then inspects equal aspect and horizontal overflow at widths 736, 360, and 320 pixels. Runs cover b=-1,0,1; zero field is expected to remain empty. Browser screenshots and timestamped console logs are kept locally in ignored evidence directories.

Issues and corrections

The in-app browser connection rejected the Linux workspace before opening a page, so verification uses a separate local Playwright Chromium session. It does not attach to a personal browser. The first integrated run passed 25 tests but failed three responsive checks because SVGRect's width/height getters were not serialized by the test harness. All slider and B-invariance assertions had already passed. Explicitly returning plain width/height properties repairs the measurement, following MDN's SVGRect documentation and Playwright's evaluation contract.

Result and learning

The browser-only rerun passed all three field-sign cases in 12.53 seconds; the subsequent full run passed 37 tests. The acceptance record lists the inspected evidence. The failure showed why a test harness needs inspection as well as the product. Independent endpoint data checks and native keyboard interaction complement the geometric measurements. We keep the browser's field formulas out of the renderer: interpolation reuses the Python-generated samples.

The only runtime network dependency is the pinned D3 script. If it cannot load, the page presents an accessible error and disables its slider instead of silently displaying an empty successful-looking chart.

Acceptance and review record

Date: 2026-10-10 (Europe/Berlin). This record describes the implemented project and the checks performed in the implementation conversation.

Verified results

- uv run pytest -q: 37 tests passed in 15.84 seconds after the mathematical, rendering, configuration, report, and browser changes. The suite generates complete artifacts independently for b=-1,0,1.

- uv run gauge-fields: successful default generation at 12:05:05 UTC. The generator's 15 gauge/strength cases use 289 grid points each; maximum absolute curl, gradient, and circulation residuals were all zero at tolerance 1e-10.

- Twenty timestamped generation/browser logs inspected at this checkpoint contained no WARNING, ERROR, or browser error entries. Earlier expected test-development failures are documented in the experiment records.

- Scientific PNG and vector PDF inspected; the figure PDF has one page and no raster images. The companion's five initial pages contain selectable experiment text and a figure appendix. This acceptance record adds a sixth page when the companion is regenerated.

- Browser behavior checked through native Home, End, and ArrowRight keys. B geometry remains unchanged at the endpoints and midpoint. The zero-field case correctly has no arrows. Layouts at 736, 360, and 320 pixels retain square plotting frames and have no horizontal overflow. Desktop and phone screenshots were inspected, as was the PDF's first rendered page.

One-pass code and documentation review

Correctness: checked the signs in A′−A, coordinate/derivative ordering, contour behavior at the scalar saddle, and the difference between a gauge parameter and physical time. The screenshot is treated as quoted source material.

Security: numeric-only serialized data is embedded in the visualization; no credentials or user-provided executable markup enter it. The repository is private, and .env, .venv, raw logs, and local test screenshots are ignored. External browser loading is limited to the version-pinned D3 script.

Maintainability and design: the thin pipeline delegates to configuration, model, verification, two renderers, and a report exporter. The browser reuses the same endpoint samples; the PDF reads existing Markdown instead of a second copy of report prose. The architecture, setup, settings, and outputs in README were checked against code and observed runs.

Reliability: configuration rejects nonfinite and unusable values, the zero field avoids normalization errors, failures preserve nonzero exit status and logged tracebacks, and a missing CDN yields a visible error state. Exact derivation and independent finite differences/line integrals provide different forms of evidence; they are not presented as interchangeable proofs.

Review corrections and limits

Git initially treated the ReportLab file as text because its initial bytes did not trigger binary detection, producing irrelevant PDF whitespace diagnostics. Explicit PDF/PNG binary attributes fix review and line-ending handling. PDF links to individual experiments now resolve to the private repository, so they remain meaningful when the PDF is downloaded separately. These are small review corrections; the prior physics and browser results still apply.

The interactive view was verified in a separate Chromium instance, not in the unavailable in-app browser connection. Other browser engines and the host's optional saved-state integration were not separately exercised. The default layout is manually inspected; parameter boundary checks do not claim exhaustive visual review of every possible configuration. Runtime dependencies, supported configuration ranges, and the CDN requirement are documented in README.

Generated field comparison

